Yuhe (Claire) Xue

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Research Interests

Programming languages
Formal methods
Compilers

Education

Sept. 2023–May 2027 **Wellesley College** | B.A., Computer Science, Minor in Mathematics
GPA: 3.68/4.00

Selected Coursework: Foundations of Computer Systems, Programming Languages, Distributed Computing, Computer Security, Theory of Computation, Deep Learning, Abstract Algebra

Candidate for Departmental Honors (Honors Thesis)

Thesis: Formally Verifying Program Optimizations via Scheduling Rewrites

Thesis Co-Advisors: Professor Adam Chlipala, Professor Alexa VanHattum

Massachusetts Institute of Technology | Cross-Registered Student
GPA: 4.80/5.00

Selected Coursework: Computer Systems Engineering, Computational Design Lab

Research Experience

June 2026–Present **MIT CSAIL**, Summer UROP and Undergraduate Thesis at PLV Group
Advisor: Professor Adam Chlipala
Extending **A Tensor Language (ATL)**, a Rocq-embedded framework for verifying semantics-preserving tensor-program scheduling rewrites, to support formal verification of loop-fusion optimizations over dependent recurrences in high-performance algorithms such as FlashAttention.
Designed and implemented **WideLetRec**, a fixed-point characterization for chains of mutually dependent, self-referencing array recurrences, and proved its correctness in representing online computations such as online softmax for verification within ATL's shallow embedding.

Jan.–June 2026 **Wellesley College**, Independent Study at Programming Systems Lab
Advisor: Professor Alexa VanHattum
Developed infrastructure toward extending the lightweight verifier Arrival for WebAssembly's production compiler Cranelift on x86-64 backends.
Extended libLISA, a Rust framework for synthesizing x86-64 instruction semantics, to accept partially specified instruction encodings with wildcard register and immediate fields, enabling synthesis over symbolic operands rather than individual concrete instructions.

Worked toward building infrastructure for translating libLISA's synthesized instruction semantics into Arrival's SMT-based specifications.

Other Experience

June–Aug. 2025 **MIT LEAP Group**, Research Software Engineer Intern
Supervisor: Software Developer Christian Gabbianelli
Built a Python - C# communication pipeline for connecting the Learning Behavior-based Knowledge Tracing model with VR psychomotor training simulations. Implemented a user's simulation progress tracker.

Publications

M. Thompson, C. Uz Bilgin, M. Huang, C. Gabbianelli, **Y. Xue**, Z. Zhao, G. Lin, S. Tadeja, and J. Liu (2026). Investigating Time and Action-Based Feedback in a VR-Based Intelligent Tutoring System for Manufacturing Psychomotor Skills. IEEE Access (under review).

Teaching Experience

Sept. 2026–Present **Foundations of Computer Systems**, Teaching Assistant
Professor Franklyn (Lyn) Turbak
Hold weekly office hours and take individual student appointments.

Jan. 2026–Present **Regression Analysis and Statistical Models**, Teaching Assistant
Professor Wendy Wang
Grade problem sets for the whole class biweekly.

Jan. 2026–June 2026 **Computing in the Age of AI**, Teaching Assistant
Professor Brian Brubach
Held weekly office hours and took individual student appointments.

Skills

Languages C, Python, Rocq, Rust, Go, Java, OCaml, C#

Tools & Technologies Linux/UNIX, Git, Z3 solver, LaTeX, Docker, AWS,

Interests Running (10k, half-marathon), Dance (Instructed at Wellesley WADO)